

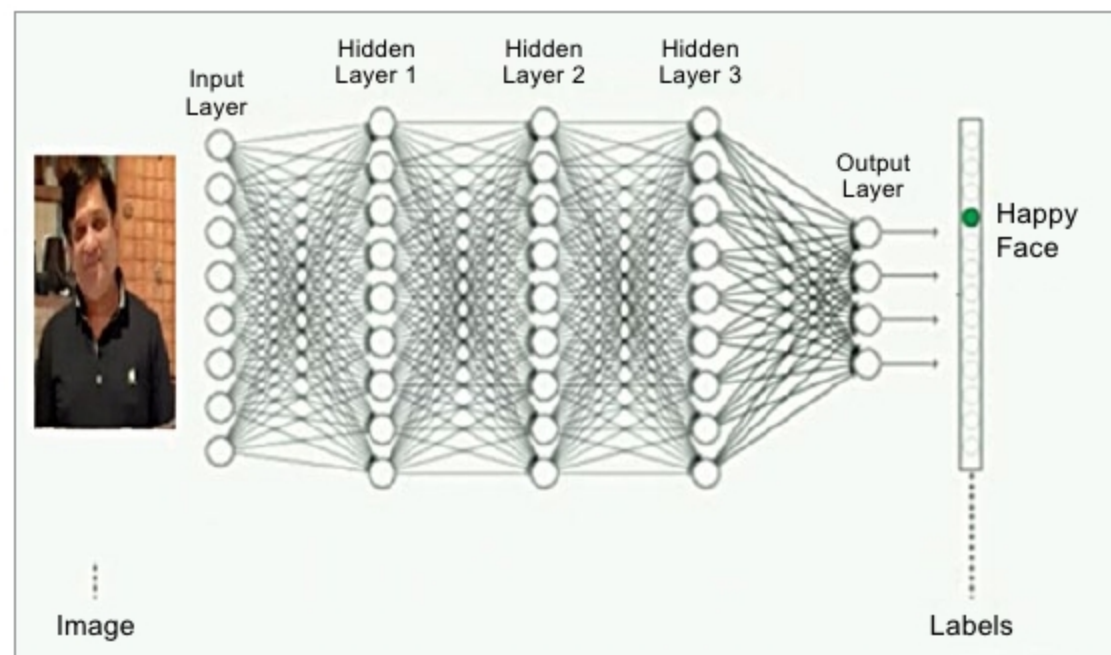


Neural Network Preceptors: How Deep Learning Can Help Solve AI Problems

This article, using an example, shows how deep learning mechanism can help solve an artificial intelligence (AI) problem. It gives a good insight on how layers in training data can be formed



Kanu Ratan Butani
is working as project manager in a software company in Mumbai. He has fifteen years of experience in Java, EAI and AI-related technologies



How deep learning mechanism reads a happy face

Creating an artificial intelligence (AI) system comes with many challenges and the biggest of all is to see how to get closer to human thinking and solve a problem the way a human will. Human or cognitive level of thinking is the most difficult, and to achieve that, it is necessary to make use of deep learning mechanism. Since convolution neural network (CNN) is the core of the deep learning mechanism, it allows adding desired intelligence to a system.

A neural network has layers of preceptors or logics/algorithms that can be written. Hence, the more layers of this logic one adds, the more the solution gets closer to cognitive thinking. The most important step here is back propagation, which helps analyse whatever has been

done in all those layers. It then provides additional help by giving three times more accurate results.

Back propagation

Let us take the classic example of the lift versus staircase. Here, apart from all other aspects, it still depends on the mood of the person—whether the person will consider taking the lift or stairs—and this can change every day.

We create infinite scenarios, yet everyday being a new day we cannot be sure what will the mood be today, and still we will try to solve this and make it as accurate as we can using the following steps.

Human mood capture. How do we capture this mood and decide what is the person's mood today? We cannot read any-